

1 Workplaces/Machines

Overview

Menu	Production Facility Management → Current Information → Workplaces/Machines
Transaction code	wpov
Function authorization	wpov

Purpose

The application *Workplaces/machines* provides an evaluation for the production management. It is intended for the following users: users from production scheduling and monitoring, schedulers, supervisors, operators or all MOC users who would like to get a comprehensive overview of the production situation at specific workplaces/machines or a complete organizational unit.

Integration

The application *Workplaces/machines* provides all kind of information that is relevant for workplaces. In addition to master data, the function also provides data required to control production processes. These are, for example:

- current workplace/machine status
- operations currently running at the workplace/machine
- currently used tools and resources
- cycle progression of the shift (for machines with clocked production)
- output per shift: quantities, durations

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the machine or workplace master data. You can also use wildcards (placeholders *).

Group from ... to ...

This selection criterion refers to the group stored in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected group. You can also use wildcards.

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows the machines matching the specified character string. You can also use wildcards (placeholders *).

Short name

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area stored in the machine master data. Note: The user can only view those machines that are included in the responsibility areas assigned to the user.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected company. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine and/or workplace master data. All workplaces/machines assigned to the selected cost center are displayed. You can also use wildcards.

Status

This selection criterion refers to the current status of machines or workplaces. All machines or workplaces, which are currently assigned to the selected status, are displayed.

Status longer than

This selection criterion refers to the current status of machines or workplaces. All machines or workplaces are shown that are currently assigned to the selected status and that are assigned to this status for a longer period than the one specified.

If several selection criteria are used, the application *Workplaces/machines* shows the results that match all selection criteria.

Detail application *Workplaces*

The detail application *Workplaces* displays all workplaces in accordance with the selections made in the selection panel. The application displays the current status, workplace information, shift quantities, cycles and number of strokes. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

In addition operation-related data is shown, if an operation is currently logged on. In case several operations are logged on, only the first operation is shown in the detail application.

Status

The *Status* column summarizes the different statuses and presents them as an "LED". The colors are as follows:

Light green	Status with RPA 11 (normally "Production")
Blue	Status with RPA 7 (normally "Setup")
Red	Status 30000 (normally "Not assigned")
Gray	Status 20000 or status with RPA 12 (normally break/no shift
Yellow	Status < 10000 and RPA <> [7,11,12] other statuses/downtimes

Master data:

Workplace

Unique ID defined in the workplace configuration.

Short name

Machine name as defined in the workplace configuration.

Designation

Long text/comment on the machine as defined in the workplace configuration.

Gruppen

Group the machine is assigned to in the workplace configuration.

Cost center

Cost center as defined in the workplace configuration.

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as defined in the workplace configuration.

Type

Workplace model according to workplace configuration.

Type

Workplace type according to the workplace configuration.

Status**Status**

Status number of the status that is currently active at the workplace. Color of the currently active status according to configuration.

Status name

Status name of the status that is currently active at the workplace.

Status since

Date when the status was assigned.

Status since

Point in time when the status was assigned.

Duration so far

Present duration of the status that is currently active at this workplace.

Predicted duration

Expected duration of the malfunction entered by the employee when assigning the status in the terminal or the duration that is stored in the status configuration.

Expected end

Calculated point in time when the malfunction ends. The calculation is based on the predicted duration. The end time is calculated using the values of *Date* + *Predicted duration*, synchronized with the Gregorian calendar.

Expected remaining runtime

Expected end minus current time, i.e. "now". If the remaining runtime is negative the expected end is already overdue. In this case, the field is highlighted in red.



Do not confuse the expected remaining runtime of the malfunction with the remaining runtime of the operation.

Shift quantities, primary quantity unit/secondary quantity unit/tertiary quantity unit/base quantity unit**Yield**

Yield that has been posted so far at the selected workplace within the current shift.

Scrap

Scrap that has been posted so far at the selected workplace in the current shift.

Rework

Rework quantity that has been posted so far at the selected workplace in the current shift.

Open quantity

Open quantity that has been posted so far at the selected workplace within the current shift.

Unit

Unit of primary quantity

Zyklus**Target cycle**

Current target cycle at the workplace.

If an operation is logged on to the machine the target cycle defined for the operation is displayed in seconds per cycle. There is no target cycle for machines to which no OP is currently logged on. In this case, "0" is entered in the "target cycle" field.

Actual cycle

Current actual cycle of the workplace

Colored display of the actual cycle relating to the configured cycle parameters.

Difference (%)

The difference in% is calculated according to the following formula: $(\text{target cycle} - \text{actual cycle}) / \text{target cycle} * 100\%$. If the actual cycle is slower than the target cycle, the difference is indicated in negative values, otherwise positive values are shown. See below for coloring.

Actual cycle (OP)

The actual cycle (OP) is a value referring to the order. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status.

Formula: $\text{Actual cycle}_{OP} = \text{RPA11}_{OP} / (\text{Yield}_{OP} / \text{Partitioning}_{OP})$

Difference (OP) (%)

The difference OP [%] column is computed according to the following formula:
 $\text{Difference}_{OP} = \text{Abs}((\text{target cycle number} - \text{actual cycle number}_{OP}) * 100) / \text{target cycle number}$

Cycle number [1/min]**Target cycle number**

1 / Target cycle

There is no target cycle for machines to which no OP is currently logged on. For this reason, the target stroke number is 0.

Actual cycle number

1 / actual cycle

Difference (%)

$(\text{target cycle number} - \text{actual cycle number}) / \text{target stroke number} * 100\%$

Please note: For rounding reasons, the difference indicated here might deviate from the difference shown in the "cycle" category.

Actual cycle number (OP)

The actual cycle number (OP) is a value relating to orders. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status. Formula: $\text{Actual cycle number}_{OP} = \text{yield}_{OP} / (\text{partitioning}_{OP} * \text{RPA11}_{OP})$

Difference (OP) (%)

The difference OP column is computed by the following formula:

$\text{Difference}_{OP} = \text{Abs}((\text{target cycle number} - \text{actual cycle number}_{OP}) / \text{target cycle number} * 100)$

Coloring of the column *Difference*

In the master data, you can define the coloring of the *Difference* column in the *Cycle* category per machine for the upper/lower action limits or upper/lower tolerance limits (menu: master data > workplaces/machines > cycle parameters). The value with a sign showing the difference is used for coloring. The value in the difference column is displayed in red if the tolerance limits are exceeded; the value is displayed in blue if the action limits are exceeded. The data is not displayed in color if no cycle parameters are defined.

Order quantities

This category displays data relating to quantities of the operation currently logged on. In case several operations are logged on, only the first operation is shown in the detail application.

Target quantity (P)

Target quantity (primary quantity unit) of the operation currently logged on.

Target scrap (P)

Target scrap (primary quantity unit) of the operation currently logged on.

Yield (P)

Posted yield (primary quantity unit) of the operation currently logged on.

Scrap (P)

Posted scrap (primary quantity unit) of the operation currently logged on.

Rework (P)

Posted rework (primary quantity unit) of the operation currently logged on.

Open quantity (P)

Posted open quantity (primary quantity unit) of the operation currently logged on.

Total quantity (P)

Total of yield + scrap + rework + outstanding quantity (open quantity)

Unit (P)

Unit of the primary quantity unit of the operation currently logged on.

Difference [%]

This difference identifies the percentage that is still to be produced to reach the target quantity of the operation. To this end, the already posted yield (P) is set in ratio to the target quantity (P):

$$\text{Difference} = 100 - (100 / \text{target quantity (P)} * \text{yield (P)})$$

The result is displayed with 2 decimal places.

Times relating to operations

This category displays the durations which are posted to the individual resource performance accounts of the logged on operation.

Detail application *Image*

The picture in the *Image* detail application shows the picture of the machine as stored in the machine configuration. The image of the machine selected in the detail application “workplace” is displayed.

The following image formats are supported: jpg, gif, png, tif, bmp, ico, emf, and wmf. The pictures have to be filed in a directory that may be accessed via the path ID “MOCWPIMG” within the path configuration.



Further information on the configuration can be found [here](#).

Detail application *Operations logged on*

The detail application *Operations logged on* shows all operations that are currently logged on to workplaces/machines which are selected in the detail application “Workplaces”. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the operation is logged on.

Order**Order**

Order number of the operation.

Sequence

Sequence number of the OP (if sequences are used).

OP

Operation number

Split

Split number of the operation (if the split function is used).

SOP

Sub operation number (reserved).

OP name/designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data.

Login**Date**

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

Primary quantity/secondary quantity/tertiary quantity/base quantity**Target quantity**

Target quantity of the operation

Unit

Unit of primary quantity

Yield

Yield that has been posted so far to the operation

Scrap

Scrap that has been posted so far to the operation

Rework

Rework quantity that has been posted so far to the operation

Open quantity

Open quantity that has been posted so far to the operation

Yield/target quantity [%]

Proportion of yield to target quantity in %

Yield since logon

Yield since the operation is logged on

Detail application *Staff logged on*

The detail application *Staff logged on* shows all persons who are logged on to the workplace selected in the detail application "Workplace". The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the operation is logged on.

Person**Name**

The person's name as defined in the HR master.

First name

The person's first name as defined in the HR master.

Name

The person's complete name as defined in the HR master (last name, middle name and first name)

Company

Company the person is assigned to in the HR master.

Personnel number

Unique key to identify the person. (Key)

Staff badge number

Staff badge number assigned to this person in the HR master.

Operator position/function

Abbreviation of the operator's function ("operator position") that has been selected when the person logged on to the machine.

Operator position/function

Unique key of the operator position that has been selected when the person logged on to this machine.

Order**Order**

Order number of the operation.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (provided that the split function is used).

SOP

Sub operation number (reserved).

OP name/designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data.

Login**Date**

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

"Advance logon" option

If this option is set, the person is logged on automatically when shifts change the next time.

Detail application *Resources logged on*

The detail application *Resources logged on* shows all resources which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the operation is logged on.

Resource**Resource type**

Resource type to which the resource is assigned.

Resource

Resource ID that is entered in the resource master data.

Designation

Resource designation recorded within master data.

Resource family

Resource family (internal ID) to which the resource is assigned.

Login**Date**

Date when the resource was last logged on to this workplace.

Time

Time when the resource was last logged on to this workplace.

Detail application *Maintenance*

The detail application *Maintenance* shows all active maintenances for the workplace that is currently selected in the selection panel. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Maintenance**Active**

Light green: Active

Status

Status of maintenance activity

Green

Blue "blue" threshold has been exceeded

Yellow "yellow" threshold has been exceeded

Red "red" threshold has been exceeded

Maintenance

Maintenance name

Type

Maintenance type defined for the maintenance:

T (cycle-based)

B (operating hours)

Z (time-based)

Class

Maintenance class

Non-recurring maintenance

Flag indicating that this maintenance is only performed once.

Valid from

Start of maintenance validity. A maintenance can only fall due within the validity period.

Valid until

End of maintenance validity.

Maintenance order

Maintenance order assigned to this maintenance.

Date

Date when this maintenance was last carried out at the selected machine.

Time

Time when this maintenance was last carried out at the selected machine.

Modified by

Person (user) who reset the last maintenance.

Actual cycles

Number of cycles accrued so far.

Next maintenance after

Counter reading of cycles when the next maintenance is to be performed.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Actual duration

Operating hours, which have been posted so far onto the resource – according to resource type settings.

Next maintenance after

Meter reading of the operating hours counter triggering the next maintenance to become due.

Interval

Interval in hours within which the maintenance is to be performed; from the maintenance configuration.

Next maintenance on

Date when the next maintenance falls due.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Info 1 - 6

Additional text 1-6 from the maintenance configuration

Detail application *Article in production*

The detail application *Article in production* shows all output materials with the relevant batch number which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace to which the batch is logged on.

Material**Material**

Material number of the currently produced article

Material designation/name

Material name of the currently produced article, which is taken over from the producing operation.

Material type

Material type of the currently produced article, which is taken over from the producing operation.

Batch number

Current batch numbers produced by the OP using this article.

Quantities**Quantities**

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Login**Date**

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who performed the last output batch change.

Detail application *Material in use*

The detail application *Material in use* shows all input materials which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the input batch is logged on.

Material**Material**

Material number of the currently logged on input batch.

Material designation/name

Material name of the currently logged on material, which is taken over from the producing operation.

Material type

Material type of the currently logged on material, which is taken over from the producing OP.

Batch number

Current batch number of the currently logged on input batch.

Quantities

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Login**Date**

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who has performed the last input batch logon.

Detail application *Status*

The detail application *Status* is only available, if the extension [rstatlist](#) is enabled.

The detail application *Status* shows the current machine status and in parallel all current resource statuses.



Parallel resource statuses refer to a workplace or a machine. The statuses do not depend on the WRM resource statuses.

The entry of other statuses than the machine status requires additional licenses and configurations.

The following paragraphs describe the data available in the table. Some data might not be displayed by default. Use the column selection function to add the required data.

Status**Status**

Number of the parallel [Resource status](#).

Status text

[Designation](#) of the parallel resource status

Status type

Number of the [Status type](#) the resource status is assigned to.

Login**Beginning of status**

Point in time when the status was set.

Duration

Duration since beginning of status. The duration is calculated based on the Gregorian calendar.

The displayed duration of the status type "MST" (machine status) can therefore differ from the value *Duration so far* in the detail application *Workplaces*.

Resource**Resource type key**

Resource type of the workplace/the machine - always "MNR"

Resource

Number of the workplace/machine

Detail application *Shift times*

The detail application *Shift times* shows RPA times of the current shift at the workplace selected in the detail application *Workplace* in a pie chart.

Detail application *Shift quantities*

The detail application *Shift quantities* shows the current shift quantities in a bar chart, i.e. yield, scrap in primary quantity unit. The quantities refer to the workplace selected in the detail application *Workplace*.

Detail application *Cycle progression*

The detail application *Cycle progression* shows the stored cycle values in a line chart in [sec/cycle]. The chart displays the cycle progression of the workplace selected in the detail application *Workplace*. By clicking a radio button the user can decide whether they want to display the current shift or the last x hours. However, x should be less than 8 hours for performance reasons.

The following limit values are displayed as lines: upper tolerance limit - UTL (red), lower tolerance limit - LTL (red), upper action limit - UAL (yellow), lower action limit - LAL (yellow). The limits are computed and displayed on the basis of the *Process parameters* configuration.

Please note: The display depends essentially on the size of the detail application.

Detail application *Downtime ranking list*

The *Downtime ranking list* shows the top x of current downtimes (status is not *production*) of the currently selected workplace during the current shift or the last hours. They are represented in a horizontal bar chart.

Using the radio buttons, it is possible to show the statuses, which have so far occurred in the current shift, or the statuses of the last x hours. By another radio button, the user can configure the display according to downtime durations or the number of respective downtimes.

The TOP X input field allows for the number of statuses to be defined (preassignment: 5).

The color of status bars corresponds to the color defined for the status text within the HYDRA configuration. The status bar is displayed in gray, in case no color is defined for the status. The status text and the value (duration in hours or number) are displayed for each bar.

Toolbar

Data collection



Log on

Use the [Log on](#) function to log on operations to the system.



Partial confirmation

Use the function "[Partial confirmation](#)" to enter part quantities for operations that are then recorded in the system.



Interrupt

Use the function "[Interrupt](#)" to interrupt operations.



Log off

Use the [Log off](#) function to log off operations.



Terminate

Interrupted or prepared operations can be logged off from the system using the [Terminate](#) function

Persons



Log person on

You can log on a person to an operation/machine using the [Log person on](#) function



Log person off

You can log off a person from the relevant operation/machine using the [Log person off](#) function